Yongda Zhu, Ph.D.
Assistant Research Professor
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RESEARCH PROFILE

I study the **phase transition** of the Universe during cosmic **reionization** and how **galaxies** and **quasars** exchange mass, metals, and ionizing radiation with the **intergalactic medium**, which contains most of the Universe's baryonic matter. I combine quasar absorption spectroscopy, **JWST**, **ALMA**, **Keck**, **Roman**, and cosmological simulations to connect galactic-scale physics with the evolution of the **cosmic baryon cycle**.

My work has constrained the **timing and topology of late reionization**, established links between IGM transmission and galaxy environment, uncovered signatures of quasar feedback on intergalactic scales, revealed **metal-enriched baryon cycling** only 500 Myr after the Big Bang, and developed **AI-enabled imaging approaches** to studying the baryon cycle with Roman and JWST.

Selected record: Ph.D. in Physics (**2024**); **17** first-author papers, including publications in *ApJ*, *ApJL*, *Nature Astronomy*, and *MNRAS*; **4,600+** total citations (h-index **40**; **530+** first-author citations); **3** student-led papers; PI/co-PI on **4** awarded Roman programs totaling **$1.2M** (**$0.6M** in direct PI support and **$0.6M** supporting collaborators through co-PI programs); PI of ALMA programs totaling **>40 hr** and more than **30 nights** on ground-based telescopes, including Keck and MMT; data or analysis leadership across approximately **2,000 hr** of competitively allocated JWST observations.

ACADEMIC APPOINTMENTS

- **Assistant Research Professor**, University of Arizona Oct. 2026 – present
  Leading research programs investigating cosmic baryon cycles with Roman and JWST.
- **JASPER Postdoctoral Scholar**, University of Arizona Aug. 2025 – Oct. 2026
  Mentors: Prof. Eiichi Egami & Prof. Xiaohui Fan
  Leading NIRCam/WFSS analysis for SAPPHIRES and studying gas–galaxy interactions across scales with JWST and machine learning.
- **Postdoctoral Research Associate**, NIRCam contract Feb. 2024 – Aug. 2025
  Mentors: Prof. Marcia Rieke & Prof. George Rieke
  JWST NIRCam Science Team and MIRI U.S. Team; led the SMILES NIRSpec data release and the JADES extended-emission search.
- **Graduate Student Researcher**, UC Riverside Jul. 2019 – Feb. 2024
  Mentor: Prof. George Becker
  Led Keck and ALMA observations of quasar sightlines showing a late, inhomogeneous end to reionization.

EDUCATION

Ph.D. in Physics, University of California, Riverside Mar. 2024
Constraining the IGM during the Later Stages of Reionization using QSO Spectra
Advisor: Prof. George Becker

M.Sc. in Physics, University of California, Riverside Dec. 2019

B.Sc. in Astronomy, University of Science and Technology of China Jun. 2018

EXTERNAL FUNDING & PI PROGRAMS

Principal-Investigator Observing Programs

- **Roman 10995** — *Establishing the Post-Reionization Lyα Transmission–Environment Baseline*. **PI: Yongda Zhu**; $150,000 awarded as direct PI research support.
- **Roman 19049** — *From Environment to Assembly: Roman Spatially Resolved Stellar Mass and SFH Maps*. **PI: Yongda Zhu**; $450,000 awarded as direct PI research support.
- **Roman 19054** — *Do Filaments Influence Galaxies Beyond Local Environment? A Roman Census of Galaxy–Filament Connectivity*. **co-PI: Yongda Zhu**; $450,000 awarded to support collaborators.
- **Roman 19029** — *Linking Galaxies to Their Neutral Gas Reservoir: A Roman–Rubin Synergy Survey of DLA Host Galaxies at Cosmic Noon*. **co-PI: Yongda Zhu**; $150,000 awarded to support collaborators.
- **ALMA 2024.1.00555.S (Cycle 11)** — *Galaxy Over/Underdensities Around IGM Transmission at $z = 5.7$: A Robust Constraint on Reionization*. **PI: Yongda Zhu**.
- **MMT/Binospec IFU 2024B, 2025B, 2026A** — Lyα escape, cold outflows, and enrichment around $z \simeq 5.5$ quasars. **PI: Yongda Zhu**.
- **ALMA 2022.1.00662.S (Cycle 9)** — *The Mean Free Path of Ionizing Photons at $z = 5.6$: A Robust Constraint on Reionization*. **PI: Yongda Zhu**.

Observing Programs Led as a Ph.D. Student at UC Riverside

UC students could not serve as Keck PIs; I led the science case and observing for each program.

- **JWST 4092** (PI: George Becker): led the science case and analysis; **Zhu** et al. 2026, ApJ, 1002, 93.
- **Keck/ESI and Keck/LRIS** (10 allocations, 2019–2024): the mean free path at $z = 5$–5.6 and long dark gaps at $z < 6$; **Zhu** et al. 2021, 2022, 2023. More than 30 ground-based nights led in total, including Keck and MMT.

Major Co-Investigator Roles

- **JWST:** co-I or data/analysis lead on ≈2,000 hr of allocated time (≈1,600 hr from the SAPPHIRES and LAPIS surveys; ≈400 hr from targeted GO programs); co-I on 13 major JWST programs, including DIVER (8018), LAPIS (10790), and the dark-gap follow-up program 7436.

- **ALMA:** co-I on six programs. **ESO/VLT:** member of Large Programme 112.262L (66 hr with MUSE).

Selected Awards and Fellowships

- OpenAI support for researchers ($12,000) 2026
- NSF/NRAO Student Observing Support Award ($40,000) 2023
- UCR Dissertation Year Fellowship ($7,200) 2022
- Benjamin C. Shen Memorial Award for Outstanding First-Year Graduate Researcher 2019
- Dean's Distinguished Fellowship, UC Riverside 2018

PUBLICATION OVERVIEW

17 first-author papers (+1 being submitted), including *Nature Astronomy*, *ApJ*, *ApJL*, and *MNRAS*; **4,600+** citations, h-index **40**, **530+** first-author citations (October 2026); full record on NASA ADS.

Selected first-author papers

- **Zhu, Y.**, Ji, Z., Becker, G. D., Ding, J., Egami, E., Fan, X., et al., **2026**. *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, *Nature Astronomy*.
- **Zhu, Y.**, Becker, G. D., D'Aloisio, A., Endsley, R., Gangolli, N., Cain, C., et al., **2026**. *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, The Astrophysical Journal, 1002, 93
- **Zhu, Y.**, Egami, E., Fan, X., Sun, F., Becker, G. D., Cain, C., et al., **2025**. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, The Astrophysical Journal, 995, L5.
- **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Cain, C., Keating, L. C., Nasir, F., et al., **2024**. *Damping wing-like features in the stacked Ly α forest: Potential neutral hydrogen islands at $z < 6$*, Monthly Notices of the Royal Astronomical Society, 533, L49.
- **Zhu, Y.**, Becker, G. D., Christenson, H. M., D'Aloisio, A., Bosman, S. E. I., Bakx, T., et al., **2023**. *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*, The Astrophysical Journal, 955, 115.
- **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., Christenson, H. M., Bañados, E., et al., **2021**. *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over $5 < z < 6$*, The Astrophysical Journal, 923, 223.

RESEARCH & COLLABORATION LEADERSHIP

- **SAPPHIRES Data Lead**: leads data reduction, validation, and delivery for a large JWST pure-parallel imaging and spectroscopy collaboration.
- **SMILES NIRSpec Lead**: leads NIRSpec data reduction, analysis, and the public high-level science-product release.
- **COSMOS-3D Large-Scale Structure Working Group Co-chair**: coordinates environmental and cosmic-web science across the collaboration.

- **Roman Science Collaboration**: member; member of the Roman WFI Calibration Working Group and participating in Roman commissioning.
- **RomanXYZ Lead Developer**: open-source reduction pipeline for Roman WFI grism spectroscopy, from count-rate images to calibrated spectra, emission-line redshifts, and line maps (public release in preparation).
- **PhotoIFU Lead Developer**: public pixel-by-pixel Bayesian SED-fitting code that turns multi-band imaging into a photometric integral-field unit.

TEACHING & MENTORING

Teaching

- **Summer Ph.D. Comprehensive-Exam Review Course**, UC Riverside: sole instructor of a month-long course for about 30 Ph.D. students (mechanics, E&M, thermodynamics, statistical mechanics) 2023
- **General Physics Laboratories** PHYS 2LA/2LB, UC Riverside: sole instructor of the lab sessions (40 students each; top-20% evaluations campus-wide) 2018–2019
- **Introductory C/C++ Programming**, USTC (teaching assistant; 160 students) 2016

Courses prepared to teach: introductory astronomy and physics; galaxies and cosmology; observational astronomy; astronomical techniques; computational astrophysics; scientific programming; and astronomical data analysis.

Research Mentoring

- **Zheng Ma**, Ph.D. student, University of Arizona 2025–present
  Two first-author papers submitted (early AGN hosts; a quasar emerging from a buried red phase); a third in preparation.
- **Junyu Zhang**, Ph.D. student, University of Arizona 2025–present
  First-author paper on a metal-poor little red dot at $z = 5.2$ (to be submitted).
- **Suprabhas Narisetty**, undergraduate, University of Arizona 2025–2026
  First-author JWST Balmer-break paper (submitted); now a Ph.D. student at UIUC.
- **Tristen Shields**, undergraduate, University of Arizona 2025–2026
  First-author JADES environment paper (to be submitted); now a Ph.D. student at Yale.
- **Seyedazim Hashemi**, Ph.D. student, UC Riverside (peer mentoring)
  First-author paper on Lyα from [O III] emitters near reionization (MNRAS, 2025).
- **Program mentoring**, UC Riverside: 14 international graduate students (2021–2023); two junior Ph.D. students (2020).

SELECTED INVITED TALKS

- Star-forming Galaxies in the Reionization Epoch, Ishigaki, Japan March 2026
- NOIRLab / Steward Observatory Joint Colloquium, **University of Arizona** February 2025
- Cosmology Seminar, **Arizona State University** February 2025
- Galaxy Seminar, **University of Michigan** November 2023

- Kashiwa-Mitaka Meeting (KMM) Seminar, **University of Tokyo** August 2023
- Lightning Talk, First Light Conference, **MIT** June 2023
- Galaxy Formation Group, **Northwestern University (CIERA)** December 2022
- FLASH Seminar, **UC Santa Cruz** November 2022
- Astronomy Lunch Talk, **UC Los Angeles** October 2022
- Astro Lunch Talk, **UC Santa Barbara** September 2022
- High-z Group Talk, **Tsinghua University** April 2022

SERVICE & OUTREACH

- **Invited grant reviewer**: NSF AST Research Grants Review Panel 2026
- **Peer reviewer**: *ApJ*, *ApJL*, *Astronomy & Astrophysics*, *JCAP*, *Open Research Europe*; ALMA and Gemini proposals
- **Local Organizing Committee**, RiekeFest 2027; built the website 2026–2027
- **Chambliss student-poster judge** and session chair, AAS 247 2026
- **Public outreach**: instructor, *Galactic Adventures*, Flandrau Science Center & Planetarium (2025); public talk, Arizona Senior Academy (2026); designed and taught K–12 courses *Multiwavelength Universe* and *Gravity Simulator* (2022); monthly Virtual Stargazing on YouTube (2020–2021).

SELECTED MEDIA COVERAGE

- "Galassie al lavoro nell'universo bambino" — **Media INAF** Oct. 2026
- "James Webb Space Telescope looks back in time, finds our ancient universe wasn't as pure as we thought" — **Space.com** Sept. 2026
- "JWST finds early galaxies were already seeding the universe with heavy elements" — **University of Arizona News** Sept. 2026
- "Supermassive serial killers: Astronomers discover how black holes 'kill off' neighboring galaxies" — **Space.com** Feb. 2026
- "Cosmic predators: How supermassive black holes slow star growth in nearby galaxies" — **University of Arizona News** Feb. 2026
- "Should I Stay or Should I Go? Clumps and Inward Migration" — **Astrobites** Feb. 2026
- "Intergalactic Impacts of Quasars During the Epoch of Reionization" — **AAS Nova** Dec. 2025
- "Astronomers discover a unique quasi-stellar object–dusty star-forming galaxy system" — **Phys.org** Nov. 2024
- "Can You Explain These Long, Dark Gaps in Your Cosmological Resume?" — **AAS Nova** Jul. 2022
- "The End of the Cosmic Dawn: Settling a Two-Decade Debate" — **SciTechDaily** Jul. 2022

YONGDA ZHU - PUBLICATION LIST

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Google Scholar: https://scholar.google.com/citations?user=wDrSZWYAAAAJ

Citation Metrics (as of October 2026):
Total citations: 4,600+ | h-index: 40 | First-author citations: 530+ | First-author papers: 17 (+1 being submitted)

First-author:

18. **Zhu, Y.**, Ma, Z., Lin, X., Zhang, J., Lyu, J., Fan, X., Egami, E., Bunker, A. J., Cai, S., Keating, L. C., Shields, J., Ji, Z., Jin, X., Liu, W., Narisetty, S., Rieke, G. H., Rieke, M. J., Rinaldi, P., and Sun, F., **2026**. *Lyα in Compact Broad-Line Emitters, Including Little Red Dots, Is More Closely Linked to the Host Galaxy than to the Broad-Line Nucleus*, arXiv e-prints, arXiv:2611.03801. Submitted to AAS journals.

17. **Zhu, Y.**, Egami, E., Rieke, M. J., Keating, L. C., Cain, C., Fan, X., Becker, G. D., Bunker, A. J., Sun, F., D'Aloisio, A., Moreno, E., D'Odorico, V., Eisenstein, D. J., Fudamoto, Y., Haehnelt, M. G., Hainline, K., Hashemi, S., Ji, Z., Kulkarni, G., Lin, X., Ma, Z., Pudoka, M., Rieke, G. H., Rinaldi, P., Robertson, B., Willmer, C. N. A., Wu, Y., and Zhang, J., **2026**. *Excess Broadband Lyman-Series Opacity in Deep JWST/NIRCam Imaging: Constraints on Galaxy-associated H I Absorption*. Submitting to *ApJ*.

16. **Zhu, Y.**, Fan, X., Keating, L. C., Becker, G. D., Egami, E., Lin, X., Sun, F., Cain, C., Rieke, M. J., Bunker, A. J., Cai, S., D'Eugenio, F., Helton, J. M., Jin, X., Li, M., Ma, Z., Maiolino, R., Rinaldi, P., Willmer, C. N. A., Wu, Y., Wu, Z., and Zhang, J., **2026**. *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over* $4.8 < z < 11$, arXiv e-prints, arXiv:2608.19311. Submitted to *ApJ*.

15. **Zhu, Y.**, Rieke, M. J., Carreira, C., Sun, Y., Ji, Z., Lyu, J., Williams, C. C., Alberts, S., Yuan, F., Zou, Y., Nakazato, Y., Bunker, A. J., Tacchella, S., Sun, F., Rieke, G. H., Egami, E., Chevallard, J., Hainline, K., Ma, Z., Pérez-González, P. G., Rinaldi, P., Rodríguez Del Pino, B., Shields, T., Stone, M., Willmer, C. N. A., Wu, Z., Yue, M., and Zhang, J., **2026**. *PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies*, arXiv e-prints, arXiv:2607.23020.

14. **Zhu, Y.**, Ji, Z., Becker, G. D., Ding, J., Egami, E., Fan, X., Jin, X., Liu, W., Lyu, J., Ma, Z., Narisetty, S., Rieke, G. H., Wu, Y., Yue, M., Zhang, J., and Rieke, M. J., **2026**. *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, *Nature Astronomy*.

13. **Zhu, Y.**, Becker, G. D., D'Aloisio, A., Endsley, R., Gangolli, N., Cain, C., Mason, C. A., Hashemi, S., and Hong, H., **2026**. *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, The Astrophysical Journal, 1002, 93

12. **Zhu, Y.**, Rieke, M. J., Ho, L. C., Sun, Y., Rieke, G. H., Yuan, F., Bakx, T. J. L. C., Becker, G. D., Yang, J., Bañados, E., Bischetti, M., Cain, C., Fan, X., Fudamoto, Y., Hashemi, S., Ikeda, R., Ji, Z., Jin, X., Liu, W., Liu, Y., Lyu, J., Ma, H.-X., Takeuchi, T. T., Umehata, H., Wang, F., and Tee, W. L., **2026**. *A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars*, The Astrophysical Journal, 1000, 312.

11. **Zhu, Y.**, Rieke, M. J., Ji, Z., Bunker, A. J., Carreira, C., Danhaive, A. L., Duan, Q., Egami, E., Eisenstein, D. J., Hainline, K., Johnson, B. D., Ma, Z., Puskás, D., Rieke, G. H., Rinaldi, P., Robertson, B., Tacchella, S., Übler, H., Villanueva, N. C., Williams, C. C., Willmer, C. N. A., Wu, Z., and Zhang, J., **2026**. *Clumps in High-redshift Galaxies: Mass Scaling and Radial Trends from JADES*, The Astrophysical Journal, 1000, 303.

10. **Zhu, Y.**, Bonaventura, N., Sun, Y., Rieke, G. H., Alberts, S., Lyu, J., Shivaei, I., Morrison, J., Ji, Z., Egami, E., Helton, J. M., Rieke, M. J., Rinaldi, P., Sun, F., and Willmer, C. N. A., **2026**. *SMILES Data Release. II. Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-resolution Spectra*, The Astrophysical Journal, 997, 301.

9. **Zhu, Y.**, Egami, E., Fan, X., Sun, F., Becker, G. D., Cain, C., Chen, H., Eilers, A.-C., Fudamoto, Y., Helton, J. M., Jin, X., Pudoka, M., Bunker, A. J., Cai, Z., Champagne, J. B., Ji, Z., Lin, X., Liu, W., Ma, H.-X., Ma, Z., Maiolino, R., Rieke, G. H., Rieke, M. J., Rinaldi, P., Sun, Y., Tee, W. L., Wang, F., Yang, J., Yue, M., and Zhang, J., **2025**. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, The Astrophysical Journal, 995, L5.

8. **Zhu, Y.**, Rieke, M. J., Ji, Z., Simmonds, C., Sun, F., Sun, Y., Alberts, S., Bhatawdekar, R., Bunker, A. J., Cargile, P. A., Carniani, S., de Graaff, A., Hainline, K., Helton, J. M., Jones, G. C., Lyu, J., Rieke, G. H., Rinaldi, P., Robertson, B., Scholtz, J., Übler, H., Williams, C. C., and Willmer, C. N. A., **2025**. *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, The Astrophysical Journal, 986, 162.

7. **Zhu, Y.**, Alberts, S., Lyu, J., Morrison, J., Rieke, G. H., Sun, Y., Helton, J. M., Ji, Z., Bhatawdekar, R., Bonaventura, N., Bunker, A. J., Lin, X., Rieke, M. J., Rinaldi, P., Shivaei, I., Willmer, C. N. A., and Zhang, J., **2025**. *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, The Astrophysical Journal, 986, 18.

6. **Zhu, Y.**, Bakx, T. J. L. C., Ikeda, R., Umehata, H., Becker, G. D., Cain, C., Champagne, J. B., Fan, X., Fudamoto, Y., Jin, X., Ma, H.-X., Sun, Y., Takeuchi, T. T., and Tee, W. L., **2024**. *Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at $z = 5.63$*, Research Notes of the American Astronomical Society, 8, 284.

5. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Cain, C., Keating, L. C., Nasir, F., D'Odorico, V., Bañados, E., Bian, F., Bischetti, M., Bolton, J. S., Chen, H., D'Aloisio, A., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Puchwein, E., Qin, Y., Ryan-Weber, E. V., Satyavolu, S., Spina, B., Walter, F., Wang, F., Wolfson, M., and Yang, J., **2024**. *Damping wing-like features in the stacked Ly α forest: Potential neutral hydrogen islands at $z < 6$*, Monthly Notices of the Royal Astronomical Society, 533, L49.

4. **Zhu, Y.**, Becker, G. D., Christenson, H. M., D'Aloisio, A., Bosman, S. E. I., Bakx, T., D'Odorico, V., Bischetti, M., Cain, C., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Haehnelt, M. G., Keating, L. C., Kulkarni, G., Lai, S., Ma, H.-X., Mesinger, A., Qin, Y., Satyavolu, S., Takeuchi, T. T., Umehata, H., and Yang, J., **2023**. *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over* $5 < z < 6$, The Astrophysical Journal, 955, 115.

3. **Zhu, Y.**, Ma, H.-X., Dong, X.-B., Huang, Y., Mistele, T., Peng, B., Long, Q., Wang, T., Chang, L., and Jin, X., **2023**. *How close dark matter haloes and MOND are to each other: three-dimensional tests based on Gaia DR2*, Monthly Notices of the Royal Astronomical Society, 519, 4479.

2. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., D'Odorico, V., Davies, R. L., Christenson, H. M., Bañados, E., Bian, F., Bischetti, M., Chen, H., Davies, F. B., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Pallottini, A., Qin, Y., Ryan-Weber, E. V., Walter, F., Wang, F., and Yang, J., **2022**. *Long Dark Gaps in the Lyβ Forest at $z < 6$: Evidence of Ultra-late Reionization from XQR-30 Spectra*, The Astrophysical Journal, 932, 76.

1. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., Christenson, H. M., Bañados, E., Bian, F., Davies, F. B., D'Odorico, V., Eilers, A.-C., Fan, X., Haehnelt, M. G., Kulkarni, G., Pallottini, A., Qin, Y., Wang, F., and Yang, J., **2021**. *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over* $5 < z < 6$, The Astrophysical Journal, 923, 223.

Student-led papers:

3. Ma, Z. (UofA grad student), **Zhu, Y.**, Ji, Z., Egami, E., Rieke, M. J., Fan, X., Lyu, J., Rieke, G. H., Sun, F., Sun, Y., Becker, G. D., Bunker, A. J., D'Eugenio, F., Jin, X., Juodžbalis, I., Liu, W., Maiolino, R., Rinaldi, P., Wang, F., Willmer, C. N. A., Wu, Y., Yang, J., Zhang, J., and Zhu, P., **2026**. *A quasar hatching from a buried red phase at $z = 3.7$*, arXiv e-prints, arXiv:2607.22966.

2. Ma, Z. (UofA grad student), Egami, E., **Zhu, Y.**, Sun, F., Lyu, J., Zhang, J., Willmer, C. N. A., Bunker, A. J., Carniani, S., Curtis-Lake, E., Hausen, R., Ji, X., Ji, Z., Juodžbalis, I., Maiolino, R., Rieke, G. H., Rinaldi, P., Sun, Y., Tacchella, S., Übler, H., and Williams, C. C., **2026**. *Undermassive Hosts of $z = 4 - 6$ AGN from JWST/NIRCam Image Decomposition with CONGRESS, FRESCO, and JADES*, arXiv e-prints, arXiv:2601.15962. Submitted to ApJ.

1. Narisetty, S. (UofA undergraduate), **Zhu, Y.**, Egami, E., Carniani, S., D'Eugenio, F., DeCoursey, C., and Willmer, C. N. A., **2026**. *Balmer Break Measurements Trace Departures from the Star-Forming Main Sequence at* $1 < z < 4$ *with JWST/NIRSpec*, arXiv e-prints, arXiv:2609.31827. Submitted to ApJ.

Co-author, major contributions (second to fourth author):

12. Ji, Z., Sun, Y., Giavalisco, M., **Zhu, Y.**, Rieke, G. H., Williams, C. C., Maseda, M. V., Lyu, J.,

Rieke, M., and Tacchella, S., **2026**. *Compact Core, Extended Reach: A Bipolar Kiloparsec-scale Elongation in a Little Red Dot at $z \approx 5.5$*, The Astrophysical Journal Letters, 1008, L55.

11. Yue, M.*, **Zhu, Y.***, Ding, J.*, Fan, X., and Luo, Y., **2026**. *A 210 Megaparsec Ionized Structure at $z = 6.8$ Revealed by an Astrophysics Foundation Model*, Research Square preprint, doi:10.21203/rs.3.rs-11272446/v1. (*Co-first authors.)

10. Ding, J., Yue, M., **Zhu, Y.**, Fan, X., and Luo, Y., **2026**. *Learning JWST. I. A Foundation Model for New Population Discoveries and Morphology-Aware Photometric Redshift Measurements in the JADES Survey*, arXiv e-prints, arXiv:2609.11879. Submitted.

9. Cain, C., Smith, B., Bowling, G. B., **Zhu, Y.**, Whitler, L., Ortiz, R., D'Aloisio, A., and Windhorst, R., **2026**. *Super-Eddington Little Blue Dots May Reionize Helium Too Early*, arXiv e-prints, arXiv:2608.28745. Submitted to ApJL.

8. Sun, Y., Ji, Z., D'Eugenio, F., **Zhu, Y.**, Rieke, G. H., Baker, W. M., Bunker, A. J., Carniani, S., Helton, J. M., Perna, M., Pérez-González, P. G., Rinaldi, P., Übler, H., and Willmer, C. N. A., **2026**. *A Census of Na D-traced neutral ISM and outflows at* $0.6 < z < 4$, arXiv e-prints, arXiv:2604.18522.

7. Hashemi, S., Becker, G. D., **Zhu, Y.**, and Hong, H., **2025**. *Lyα emission from [O III] emitters near reionization: the role of environment in galaxy Lyα detection*, Monthly Notices of the Royal Astronomical Society, 542, 104.

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